

VR-Forms. Preprint.

Part VI. Forms of Speech

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VI.0. Introduction

Part V applied the formal apparatus to classical mathematics: classical $\mathbb{R}$, $\wp(\mathbb{N})$, Vitali, Russell's class, proper classes — all received the status of formal terms. Part VI extends the application beyond mathematics: to mythological beings, theological concepts, literary figures, philosophical categories.

This is not a rhetorical move and not an artistic extension. It is a **test of the universality** of the formal register. If the formal register really covers speech without exception, it must work uniformly with mathematical and non-mathematical descriptions. Part VI demonstrates exactly this.

Contents: universality of the formal register (§VI.1); mythological terms (§VI.2); theological terms (§VI.3); philosophical categories as formal terms (§VI.4); literary figures (§VI.5); status of non-mathematical terms and its philosophical significance (§VI.6); summary (§VI.7).

VI.1. Universality of the formal register

The principle of forms (§II.4) states: every syntactically correct record of a description specifies a formal term. No restrictions on the content of the description are imposed. The description may be mathematical, may be non-mathematical, may refer to the existing, the non-existing, the imagined, the paradoxical. The principle of forms applies to all uniformly.

This means: the formal register of VR-Forms is a **universal language of descriptions**. Everything for which a correct syntactic description can be formed has a place in it as a formal term.

Remark on syntactic correctness

«Syntactic correctness» in VR-Forms is a formal notion, set by the rules of §II.1. A description is correct if it is built by these rules. This means the syntax of VR-Forms is a fixed formal language; descriptions in natural language («a fire-breathing dragon», «an omnipotent god») are not directly expressed in it.

However, every meaningful natural-language description can be translated into the formal — if a definite predicate is assigned to it. «Dragon» becomes a formal term $\ulcorner \{x : x \text{ is a fire-breathing lizard-like creature with wings} \} \urcorner$, where «is a fire-breathing lizard-like creature with wings» is a predicate expressed in the language of VR-Forms (via a chosen system of feature designations). Similarly with «god», «soul», «unicorn».

Thus the non-mathematical terms in VR-Forms are not literary figures as such, but **their formalisations** through predicates. The choice of formalisation depends on context; different contexts may give different «dragons» (Chinese, European, mythological, literary), and each formalisation is its own formal term.

VI.2. Mythological terms

Dragon

「dragon」 is a formal term. Formalisation: 「 $\{x : \text{fire_breathing}(x) \wedge \text{lizard_like}(x) \wedge \text{winged}(x)\}$ 」, where the predicates «fire_breathing», «lizard_like», «winged» are elements of the chosen feature system.

In the ontological register of VR-Sets: the corresponding operational set is absent. The description «a functionality enumerating all fire-breathing lizard-like winged creatures» is not a procedure over $\emptyset$; no operational action realises it.

In the formal register: 「dragon」 is a formal term. One can speak of it, formulate statements: «「dragon」 breathes fire» (true in the formal register by definition of 「dragon」), «「dragon」 exists» — a statement of the formal register, having no ontological force.

Unicorn, phoenix, centaur, minotaur

Analogously. Each is a formal term with its own formalisation:

「unicorn」 = 「 $\{x : \text{horse_like}(x) \wedge \text{single_horned}(x)\}$ 」

「phoenix」 = 「 $\{x : \text{bird_like}(x) \wedge \text{reborn_from_ashes}(x)\}$ 」

「centaur」 = 「 $\{x : \text{has_human_upper_body}(x) \wedge \text{has_horse_lower_body}(x)\}$ 」

All formal terms. None operationally realisable. For each, one can formulate correct statements in the formal register that follow from the definition.

Remark on biological consistency

Some mythological terms are formally **contradictory** in a strong sense — that is, their definitions rely on properties biologically incompatible. For example, 「centaur」 with two cardiovascular systems, or 「dragon」 with an endogenous fire organ without thermal damage.

This does not make the formal term inadmissible in VR-Forms. The principle of forms does not require biological or physical consistency. Syntactic correctness suffices.

This is an analogue of the mathematical case in which contradictory formal terms are admissible (Russell's class). In the formal register contradictoriness is a property of the description, not a prohibition.

VI.3. Theological terms

God in monotheistic understanding

$\ulcorner \text{god} \urcorner = \ulcorner \{x : \text{unique}(x) \wedge \text{omnipotent}(x) \wedge \text{eternal}(x) \wedge \text{creator_of_world}(x)\} \urcorner$ — a formal term. The formalisation is the standard theological definition translated into predicate form.

In the ontological register of VR-Sets: the corresponding operational set is absent. No procedure over $\emptyset$ realises «the unique omnipotent eternal creator». Not as a «prohibition» but as a description signifying nothing in the operational ontology: operationally not reproducible.

In the formal register: $\ulcorner \text{god} \urcorner$ is a formal term. One can speak of it, prove statements following from the definition: $\ulcorner \text{god} \urcorner$ is unique (by definition), $\ulcorner \text{god} \urcorner$ created $\ulcorner \text{the world} \urcorner$ (if $\ulcorner \text{the world} \urcorner$ is a separate formal term), and so on.

Classical theological paradoxes

The paradox of the stone (can an omnipotent god create a stone he cannot lift?) in VR-Forms is treated as a statement of the formal register about the properties of $\ulcorner \text{god} \urcorner$, in particular about the consistency of the predicate «omnipotent». If the formalisation of omnipotence is contradictory (as in this paradox), then $\ulcorner \text{god} \urcorner$ in the given formalisation contains a contradictory characteristic — analogously to the Russell term. This is a property of the **description**, not an ontological failure.

This reinterpretation is consistent with the general position of VR-Forms: paradoxicality of theological concepts is a property of the chosen formalisation. Other formalisations (omnipotence as «can do all that is logically possible») generate no such contradiction; and these other formalisations are their own formal terms.

God in polytheistic and other systems

$\ulcorner \text{Zeus} \urcorner$, $\ulcorner \text{Odin} \urcorner$, $\ulcorner \text{Brahma} \urcorner$, $\ulcorner \text{Allah} \urcorner$ — each a formal term with its own formalisation. Each description specifies its own formal term. Statements about them belong to the formal register.

VR-Forms neither asserts nor denies the ontological reality of these entities. It only states: in the VR-ontology they are not (for the same reasons as unicorns: no operational action over $\emptyset$). In the formal register they are admissible as formal terms. About their existence **outside** the VR-ontology the present work does not speak.

VI.4. Philosophical categories

Classical philosophical categories — universals, essence, accident, form, matter, soul, mind — are formal terms in VR-Forms by the same principle: each description, regardless of its philosophical tradition, satisfies syntactic correctness and so specifies a formal term.

Universals

«Universal» in the classical philosophical tradition — a common nature shared by many individuals. 「universal」 is a formal term. Each concrete instance (「redness」, 「humanity」, 「triangularity」) is its own formal term with its own formalisation.

The medieval dispute of realists and nominalists is a dispute about whether universals have ontological correlates. In terms of VR-Forms: whether formal terms like 「redness」 have an operational realisation.

The position of VR-Forms is closer to nominalism: in its ontology ($\emptyset$ + the operational) universals as self-standing objects are absent. They are admissible as formal terms — that is, as names, not entities. This is a precise formulation of medieval nominalism in the modern formal apparatus.

Aristotelian and scholastic categories

「essence」, 「accident」, 「form」, 「matter」 — all formal terms, with formalisations depending on the chosen scholastic system (Aristotle, Aquinas, Suárez each give different formalisations). VR-Forms does not adjudicate classical metaphysical disputes; it states only that these categories are formal terms, work with them belongs to the formal register, and their operational realisability in VR-Sets is absent.

Soul, spirit, mind

「soul」, 「spirit」, 「mind」 — formal terms. Whether 「mind」 can be reduced to the operational (for instance, to a procedure over a physically realised neural network) is a separate question — about the existence of an operational realisation of the corresponding formal term. VR-Forms does not answer it; it provides a **language** for formulating it.

VI.5. Literary figures

Literary characters and worlds form a particularly rich domain of formal terms.

Characters and worlds

Literary characters (「Hamlet」, 「Sherlock Holmes」, 「Anna Karenina」) are formal terms. Each carries an extensive formalisation, set by the corresponding literary work.

Literary worlds (「Middle-earth」, 「Hogwarts」, 「Narnia」) are also formal terms, often containing their own formal terms (「elf」, 「hobbit」, 「Harry's wand」) — nested structures of formal terms.

Truth in literature

The statement «Sherlock Holmes lived on Baker Street» — true in what sense? In VR-Forms this is a formal-register statement about the formal term 「Sherlock Holmes」. Truth is determined by the chosen formalisation (the text of the work as a source of predicates).

«Sherlock Holmes exists» in VR-Forms: in the formal register admissible ($\ulcorner$ Sherlock Holmes $\urcorner$ is a formal term, in this sense «exists»); in the ontological register the operational correlate is absent.

This two-register treatment corresponds to the intuitive distinction «exists in the work» vs «exists in reality», but gives it a formal form.

VI.6. Status of non-mathematical terms: philosophical significance

Let us list what has been shown in Part VI, and what conclusions follow.

(1) The formal register is universal

Demonstrated: the formal apparatus of VR-Forms works with mathematical (Part V) and non-mathematical (the present part) descriptions uniformly. The principle of forms and the transit rule have no restriction by subject matter.

This asserts the universal character of the two-register structure: the division into operational and formal applies to speech in general, not specifically to mathematics.

(2) Ontological status of non-mathematical terms

In the VR-ontology non-mathematical formal terms (dragons, gods, unicorns) have no operational correlate. This is **not an ontological denial**. It is a statement: in the VR-ontology ($\emptyset$ + the operational) they do not exist. Whether they exist in another ontology (theological, literary, psychological) is a separate question, not pertaining to VR-Forms.

The distinction «not in the VR-ontology» and «not at all» is principal. VR-Forms does not claim exclusivity for its ontology. It only clearly fixes it, and relative to it classifies formal terms.

(3) Connection with nominalism

VR-Forms gives a precise formal form to the classical nominalist position: universals and general entities are names (formal terms), only the concrete (the operational) is real. Difference from classical nominalism: VR-Forms replaces «the concrete» with «the operational» (what in VR has an operational correlate).

This is a consistent extension of nominalism from objects to operations: real are not «concrete objects» (which in modern physics is already questionable) but operational actions upon the fundamental $\emptyset$. Names — everything else, including «humanity», «redness», «Socrates as a man» in the universal sense.

(4) Substantive role of the formal register in non-mathematical speech

What practical significance does the formal register have for non-mathematical descriptions?

Above all, **methodological**. Statements about $\ulcorner$ dragon $\urcorner$, $\ulcorner$ god $\urcorner$, $\ulcorner$ soul $\urcorner$ can be formulated and discussed formally, without claiming their ontological reality. This removes a classical

difficulty: «is there sense in talking about the non-existent?» — answered by: the formal register provides meaningful speech without ontological commitments.

This is relevant to theology (one may build theological systems as formal structures without making ontological claims), literary studies (literary worlds as formal structures), philosophy (categories as formal terms), mythology (mythological systems as formal systems).

VR-Forms does not take away from these areas their content. It provides them with an **apparatus** for talking about them, clearly separating the internal logic of the system from the question of its ontological reality.

VI.7. Summary of Part VI

Established:

- (1) The formal register of VR-Forms is universal: it covers mathematical and non-mathematical descriptions uniformly (§VI.1).
- (2) Mythological terms (dragons, unicorns, phoenixes) are formal terms, not operationally realisable in VR (§VI.2).
- (3) Theological terms (god, divine entities) are formal terms; classical theological paradoxes are treated as properties of the formalisation (§VI.3).
- (4) Philosophical categories (universals, essence, soul) are formal terms; VR-Forms provides a formal form for classical nominalism (§VI.4).
- (5) Literary figures and worlds are formal terms; «truth in the work» receives a precise formal reading (§VI.5).
- (6) Philosophical significance of the universality of the formal register: ontological clarity relative to the VR-ontology, without a claim of exclusivity; methodological usefulness for theology, literary studies, philosophy (§VI.6).

Substantive summary. VR-Forms is not only a formal apparatus for mathematical foundations. It is a **universal language of speech about the non-operational**. Dragons and classical $\mathbb{R}$ have in it the same status: formal terms, legitimate in the formal register, having no operational correlate in the VR-ontology. This is the position that in Part I was stated verbally; the present part demonstrates it in operation.

What follows. Part VII is devoted to the logic of the two-register system — what inference rules act in each register, how they interact, whether classical logic is preserved. This is a technical closure of the system.